Foundations of Mixture model

[Mixture model]

1.0 Content Block

The component model parameters θi are also calculated by expectation maximization using data points xj that have been weighted using the membership values. For example, if θ is a mean μ

2.0 Content Block

θ i = 1 … K : {\displaystyle {\boldsymbol {\theta }}_{i=1\dots K}:} vector of dimension V , {\displaystyle V,} composed of θ i , 1 … V ; {\displaystyle \theta _{i,1\dots V};} must sum to 1

3.0 Content Block

In image processing and computer vision, traditional image segmentation models often assign to one pixel only one exclusive pattern. In fuzzy or soft segmentation, any pattern can have certain "ownership" over any single pixel. If the patterns are Gaussian, fuzzy segmentation naturally results in Gaussian mixtures. Combined with other analytic or geometric tools (e.g., phase transitions over diffusive boundaries), such spatially regularized mixture models could lead to more realistic and computationally efficient segmentation methods.

4.0 Content Block

μ s (j + 1) = $\sum$ t = 1 N h s (j) (t) x (t) $\sum$ t = 1 N h s (j) (t) {\displaystyle \mu _{s}^{(j+1)}={\frac {\sum _{t=1}^{N}h_{s}^{(j)}(t)x^{(t)}}{\sum _{t=1}^{N}h_{s}^{(j)}(t)}}}

5.0 Content Block

A prior distribution is placed over the parameters describing the topic distributions, using a Dirichlet distribution with a concentration parameter that is set significantly below 1, so as to encourage sparse distributions (where only a small number of words have significantly non-zero probabilities). Some sort of additional constraint is placed over the topic identities of words, to take advantage of natural clustering. For example, a Markov chain could be placed on the topic identities (i.e., the latent variables specifying the mixture component of each observation), corresponding to the fact that nearby words belong to similar topics. (This results in a hidden Markov model, specifically one where a prior distribution is placed over state transitions that favors transitions that stay in the same state.) Another possibility is the latent Dirichlet allocation model, which divides up the words into D different documents and assumes that in each document only a small number of topics occur with any frequency.

6.0 Content Block

K , N = as above φ i = 1 … K , φ = as above z i = 1 … N , x i = 1 … N = as above θ i = 1 … K = { μ i = 1 … K , σ i = 1 … K 2 } μ i = 1 … K = mean of component i σ i = 1 … K 2 = variance of component i μ 0 , λ , ν , σ 0 2 = shared hyperparameters μ i = 1 … K ~ N (μ 0 , λ σ i 2) σ i = 1 … K 2 ~ I n v e r s e - G a m m a ■ (ν , σ 0 2) φ ~ S y m m e t r i c - D i r i c h l e t K ■ (β) z i = 1 … N ~ Categorical ■ (φ) x i = 1 … N ~ N (μ z i , σ z i 2) {\displaystyle {\begin{array}{lcl}K,N&=&{\text{as above}}\\\phi _{i=1\dots K},{\boldsymbol {\phi }}&=&{\text{as above}}\\z_{i=1\dots N},x_{i=1\dots N}&=&{\text{as above}}\\\theta _{i=1\dots K}&=&\{\mu _{i=1\dots K},\sigma _{i=1\dots K}^{2}\}\\\mu _{i=1\dots K}&=&{\text{mean of component }}i\\\sigma _{i=1\dots K}^{2}&=&{\text{variance of component }}i\\\mu _{0},\lambda ,\nu ,\sigma _{0}^{2}&=&{\text{shared hyperparameters}}\\\mu _{i=1\dots K}&\sim &{\mathcal {N}}(\mu _{0},\lambda \sigma _{i}^{2})\\\sigma _{i=1\dots K}^{2}&\sim &\operatorname {Inverse-Gamma} (\nu ,\sigma _{0}^{2})\\{\boldsymbol {\phi }}&\sim &\operatorname {Symmetric-Dirichlet} _{K}(\beta)\\z_{i=1\dots

N}&\sim &\operatorname {Categorical} ({\boldsymbol {\phi }})\\x_{i=1\dots N}&\sim &{\mathcal {N}}(\mu _{z_{i}},\sigma _{z_{i}}^{2})\end{array}}}

7.0 Content Block

Multivariate Gaussian mixture model A Bayesian Gaussian mixture model is commonly extended to fit a vector of unknown parameters (denoted in bold), or multivariate normal distributions. In a multivariate distribution (i.e. one modelling a vector x {\displaystyle {\boldsymbol {x}}} with N random variables) one may model a vector of parameters (such as several observations of a signal or patches within an image) using a Gaussian mixture model prior distribution on the vector of estimates given by

8.0 Content Block

Point set registration Probabilistic mixture models such as Gaussian mixture models (GMM) are used to resolve point set registration problems in image processing and computer vision fields. For pair-wise point set registration, one point set is regarded as the centroids of mixture models, and the other point set is regarded as data points (observations). State-of-the-art methods are e.g. coherent point drift (CPD) and Student's t-distribution mixture models (TMM). The result of recent research demonstrate the superiority of hybrid mixture models (e.g. combining Student's t-distribution and Watson distribution/Bingham distribution to model spatial positions and axes orientations separately) compare to CPD and TMM, in terms of inherent robustness, accuracy and discriminative capacity.

9.0 Content Block

K = { p (·) : p (·) = ∑ i = 1 n a i f i (· ; θ i) , a i > 0 , ∑ i = 1 n a i = 1 , f i (· ; θ i) ∈ J ∀ i , n } {\displaystyle K=\left\{p(\cdot):p(\cdot)=\sum _{i=1}^{n}a_{i}f_{i}(\cdot ;\theta _{i}),a_{i}>0,\sum _{i=1}^{n}a_{i}=1,f_{i}(\cdot ;\theta _{i})\in J\ \forall i,n\right\}}

10.0 Content Block

Expectation maximization (EM) Expectation maximization (EM) is seemingly the most popular technique used to determine the parameters of a mixture with an a priori given number of components. This is a particular way of implementing maximum likelihood estimation for this problem. EM is of particular appeal for finite normal mixtures where closed-form expressions are possible such as in the following iterative algorithm by Dempster et al. (1977)

11.0 Content Block

p (θ) = ∑ i = 1 K φ i N (μ i , Σ i) {\displaystyle p({\boldsymbol {\theta }})=\sum _{i=1}^{K}\phi _{i}{\mathcal {N}}({\boldsymbol {\mu }}_{i},{\boldsymbol {\Sigma }}_{i})}

12.0 Content Block

Financial returns often behave differently in normal situations and during crisis times. A mixture model for return data seems reasonable. Sometimes the model used is a jump-diffusion model, or as a mixture of two normal distributions. See Financial economics § Challenges and criticism and Financial risk management § Banking for further context.